

CN Assignment # 3

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Question P3

Step	$D(t), p(t)$	$D(u), p(u)$	$D(v), p(v)$	$D(w), p(w)$	$D(y), p(y)$	$D(z), p(z)$	
	N'	∞	∞	<u>3, n</u>	6, n	6, n	8, n
0	x						
1	xv	7, v	<u>6, v</u>	3, n	6, n	6, n	8, n
2	xvu	7, v	6, v	<u>3, n</u>	<u>6, n</u>	6, n	8, n
3	xvuw	7, v	6, v	3, n	6, n	<u>6, n</u>	8, n
4	xvuwyt	<u>7, v</u>	6, v	3, n	6, n	6, n	8, n
5	xvuwytz	7, v	6, v	3, n	6, n	6, n	<u>8, n</u>
6	xvuwytz	7, v	6, v	3, n	6, n	6, n	8, n

Showing ^{step} 1 equations same for all steps,

Step 1: Add v (smallest cost node 3)

Update distances via v:

- $D(u) = \min(\infty, D(v) + 3) = 6, p(u) = v$
- $D(w) = \min(6, D(w) + 4) = 6, p(w) = \cancel{x}$ (no change)
- $D(t) = \min(\infty, D(t) + 4) = 7, p(t) = v$
- $D(y) = \min(6, D(y) + 8) = 6, p(y) = x$ (no change)

Question P(5)

		Cost to					
		u	v	x	y	z	
From	v	∞	∞	∞	∞	∞	
	x	∞	∞	∞	∞	∞	
	z	∞	6	2	∞	0	

Iteration 1

From v: $D_v(u) = 1$, $D_v(v) = 0$, $D_v(x) = 3$, $D_v(y) = \infty$, $D_v(z) = 6$

From x: $D_x(u) = \infty$, $D_x(v) = 3$, $D_x(x) = 0$, $D_x(y) = 3$, $D_x(z) = 2$

Updates at z

- $D_z(u) = \min(\infty, c(z, v) + D_v(u)) = 6 + 1 = 7$
- $D_z(v) = \min(7, c(z, x) + D_x(v)) = 2 + 3 = 5$
- $D_z(x) = 2$ (no change)
- $D_z(y) = \min(\infty, c(z, x) + D_x(y)) = 2 + 3 = 5$

		Cost to					
		u	v	x	y	z	
From	v	1	0	3	∞	6	
	x	∞	3	0	3	2	
	z	7	5	2	5	0	

Iteration 2

From v: $D_v(y) = 3$, so $D_z(y) = \min(5, c(z, v) + D_v(y)) = 6 + 3 = 9$ (no change).

From x: No significant updates, as difference remains constant.

		Cost to					
		u	v	x	y	z	
From	v	1	0	3	3	5	
	x	4	3	0	3	2	
	z	6	5	2	5	0	

Iteration 3

After further updates, the table stabilizes, and no further changes occur. The final state reflects the converged shortest distances.

		Cost to				
		U	V	X	Y	Z
From	V	1	0	3	3	5
	X	4	3	0	3	2
	Z	6	5	2	5	0

Question P9

1. Decreasing the Cost of a Link

No, decreasing the cost of a link improves the path between two nodes and does not lead to routing loops. Since distance-vector updates propagate the reduced cost, nodes will update their routing tables to reflect the shorter path. This process converges quickly and does not involve cyclic updates or inconsistencies.

2. Connecting Two Nodes Without a Previous Link

No, connecting two nodes with a new link is equivalent to decreasing the link cost from infinity to a finite loop. This change introduces a new, shorter path for nodes, leading to updates that improve routing. Like the first case, this does not create routing loops or cyclic updates, as it does not involve any next-hop dependency cycles that would trigger the count-to-infinity issue.

Question P14

- (a)
- Router 3c is part of AS3.
 - It learns about prefix x from AS1 or AS4 through eBGP, as BGP is the protocol used for routing between different ASes.
 - **eBGP** operates at the borders of AS3 and exchanges routing information with AS1 or AS4.
- (b)
- Router 3a is also part of AS3.
 - Within an AS, routers propagate routes learned from eBGP using iBGP.
 - Router 3a learns about x from router 3c (or another iBGP peer in AS3) through **iBGP**.
- (c)
- Router 1c is part of AS1.
 - Prefix x originates in AS3 or AS4 and is shared with AS1 via **eBGP** at the border routers.

Question P15

- (a)
- Router 1d selects the interface I1 because it leads to the least cost path toward the gateway router 1c, which advertises the route to x .
 - OSPF or RIP computes the shortest path within the AS, and I1 is determined as the optimal interface.
 - Answer: **I1**

cb) Answer: I2

- With AS2 and AS4 directly connected, router 1d learns two routes to x: one via AS3 and one via AS2.
- Both paths have equal AS - PATH length, but router 1d chooses I2, as it leads to the closest NEXT - HOP router (shorter intra - AS path).

cc) Answer: I1

- Now, the route via AS2 includes an additional hop through AS5, making the AS - PATH for this route longer.
- Router 1d prefers I1, as it leads to the Shorter AS - PATH via AS3.

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